MD. SHAHARIAR HOSSAIN

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LinkedIn | Personal Website | ResearchGate | Google Scholar

EDUCATION

M.Sc. in Data Science — **Current CGPA: 4.00 / 4.00**
BRAC University, Dhaka, Bangladesh — *Expected Graduation: May 2027*

B.Sc. in Electrical and Electronic Engineering — **CGPA: 3.60 / 4.00**
BRAC University, Dhaka, Bangladesh — *May 2026*

PUBLICATIONS

Journal:

1. **M. S. Hossain**, M. T. Hassan and A. Islam " **G-buffer-guided feature modulation for lightweight rendered image super-resolution**," *Array*, Volume 32, 2026, 101324, ISSN 2590-0056, doi:10.1016/j.array.2026.101324.

Conference Paper:

1. **M. S. Hossain**, R. Abrar, S. A. Akib, T. Ahmed, S. Alam and A. M. Oni. "**Kinematic Analysis, Trajectory Generation, and Hardware Validation for Quadruped Robot Leg**," *IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)*, (Accepted, 2026)

2. **M. S. Hossain**, M. T. Hassan and M. S. Hossain, "**Biologically Inspired Network Design to Train Only Through Forward Pass**," *2024 2nd International Conference on Information and Communication Technology (ICICT)*, Dhaka, Bangladesh, 2024, pp. 130–134, doi: 10.1109/ICICT64387.2024.10839663.

3. **M. S. Hossain**, M. S. Hossain and M. B. Karim, "**Training StyleGAN2 on BWW-Texture Dataset: Generating Textures Through Transfer Learning**," *2023 International Conference on Information and Communication Technology for Sustainable Development (ICICT4SD)*, Dhaka, Bangladesh, 2023, pp. 341–345, doi: 10.1109/ICICT4SD59951.2023.10303468.

Book:

1. M. S. Hossain and **M. S. Hossain** (co-author), "**Mohabishwer Mohajatra**" [The Great Journey of the Universe], ISBN: 978 984-95421-7-9, A 284-page science book on cosmology and the history of the universe. (1st edition: March, 2021, 2nd print: February 2023.)

RESEARCH EXPERIENCE

1. **Research Assistant, BSRM SoE** (September 2026 – Present, BRAC University)
   Role: **Autonomous Systems, Swarm Control, State-Space Modeling, & Edge Robotics** — *Supervisor:* **Md Arif Billah, Ph.D**, Assistant Professor, EEE

   - Derived theoretical **state-space models**, feedback control equations, and dynamic block diagrams for **swarm robotics**; validated non-linear system dynamics using **MATLAB** graphical simulations.
   - Designed 3D robot models in **Autodesk Fusion** and simulations for complex environment testing.
   - Developed a new swarm platform of **5 robotic agents** built with economical and **efficient local edge-hardware**.
   - Implemented **Reinforcement Learning (RL)** algorithms and AI-driven **world models** utilizing latent state representations for robust, adaptive trajectory execution and autonomous agent control.

2. **Directed Research, EEE498** (June 2025 – May 2026, BRAC University)
   Project: **G-Buffer-Guided Feature Modulation for Lightweight Rendered Image Super-Resolution** — *Supervisor:* **Amirul Islam, Ph.D**, Associate Professor, EEE

- Designed lightweight **FiLM, Attention, Convolution** architectures in TensorFlow fusing a low-resolution render with high-resolution auxiliary buffer to reconstruct a **2,3,4× upscaled image**.
- Built a custom paired **dataset (10,000+ frame and auxiliary data pairs)** from rendering pipelines.
- Implemented **ablation studies + retraining** on fusion of auxiliary channels; **PSNR / SSIM** evaluation.

3. **Final Year Design Project, EEE 499** (June 2025 – May 2026 , BRAC University)

Project: **A Quadruped Robot for Real Time Inspection and Responsive Operations.** *Advisory Panel:* **Academic Technical Committee**, EEE

- Built a **quadrupedal robot platform** for remote inspection in confined, high-risk industrial sites (nuclear, oil & gas, chemical, construction), with **onboard perception** to reduce human exposure.
- Developed custom **Reinforcement Learning**, **real-time perception**, **Inverse Kinematics**, **trot and walking gait** for quadruped locomotion.
- Designed 3D robot models in Autodesk Fusion and developed high-fidelity physical hardware simulations in **ROS/Gazebo** and **NVIDIA Isaac Sim** for complex environment testing.
- **Coordinated a 4-member team** and presented iterative design reviews to faculty and industry partners; **led safety analysis** and integrated **ISO 45001** standards into design.

TEST SCORES

IELTS: Overall Band **7.5** *(Taken 1st Nov 2026)*
Listening **8.5** – Reading **7.5** – Writing **6.5** – Speaking **7.0**

HONORS & AWARDS

- **Special Recognition Awarded**, Certificate for exceptional extracurricular achievements, BRAC University Freshers Orientation Program (Sep 2026).
- **Champion** (1st Place), Project Showcase (A Quadruped Robot for Real-Time Inspection and Responsive Operations), East-West National Robo Fest (June 2026).
- **Champion** (1st Place), Bangladesh Physics Olympiad; Rajshahi Regional Round; National Finalist (2016).
- **First Prize**, BCSIR National Science Fair ("Electricity-Generating Road"), Project Leader (Feb 2020).
- **National Olympiad Finalist (Bangladesh):** Astrophysics (2016–2021) [Top-5]; Physics (2020) [Top-10]; Junior Science (2017) [Top-20]; National High School Programming Contest (2016) [Top-20].
- **Best Creative Writer (Science Division)**, National Creative Writing Competition "Jatiya Shrijanshil Medha Annyashan" (2020).

LEADERSHIP & OUTREACH

- **Member**, IEEE Student Branch and Robotics Club, BRAC University (2022–Present): ML / vision demos, embedded systems workshops, and robotics outreach for new undergraduates.
- **Co-Founder**, "Science Galaxy" STEM club (Rajshahi University School & College): organized science fairs, talks, and student research showcases (2016–2020).
- **Mentor** for Bangladesh Physics / Math / Astrophysics Olympiads high school training workshops (2018–2020).

TECHNICAL SKILLS

- **Programming & ML:** Python (TensorFlow, PyTorch, NumPy), C/C++.
- **Computer-Aided Design:** Fusion360, Altium, AutoCAD, Adobe Illustrator, Blender, Unreal Engine.
- **Simulation Software:** NVIDIA Isaac Sim, Webots, PSpice, Proteus, MATLAB.
- **Productivity Software:** MS Office, LaTeX.
- **Tools:** Linux, Arduino, Raspberry Pi, ATmega32.